

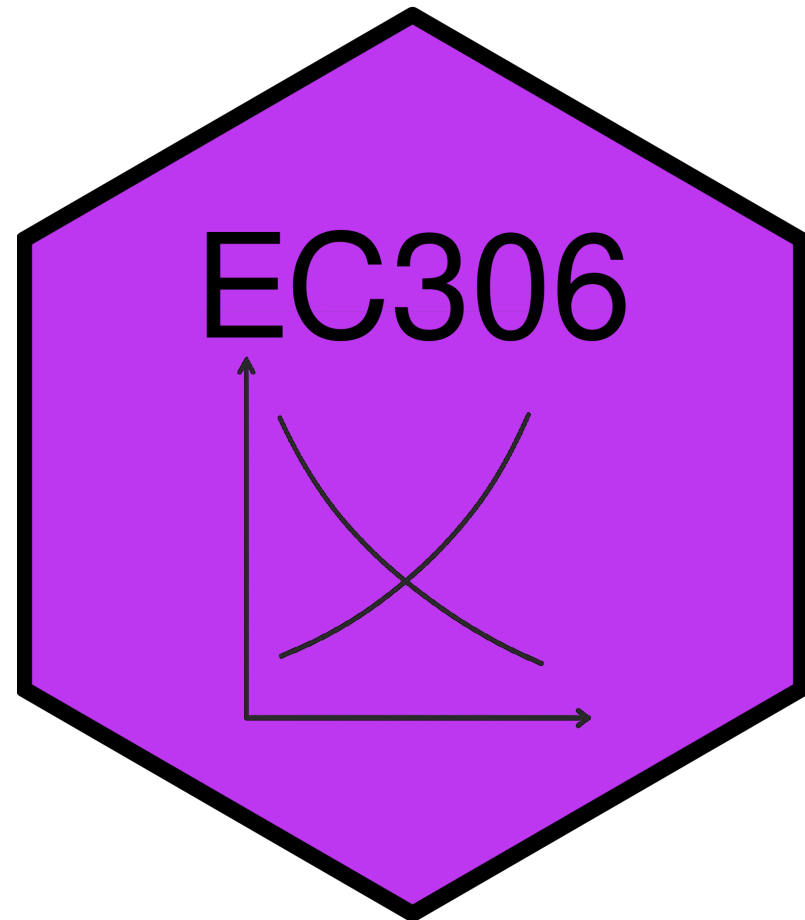
Imperfectly Competitive Labour Markets

EC306- Wages and Employment

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Goals of This Section

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In this section we will:

- Discuss the effect of market power in the labour market on equilibrium wages and employment
- Derive the equilibrium in a monopsonistic labour market
- Examine the evidence for monopsony power
- Analyze minimum wages in competitive and monopsonistic markets

Monopsony in the Labour Market

Introduction

! Monopsony

Monopsony: a single buyer with market power in the labour market

- Contrast with monopoly: single seller with market power in the goods market

Some examples of monopsony in the labour market:

- Company/factory towns
- Hospital labour markets
- University professor labour markets
- Public sector in small areas

In all cases, there is a large buyer of labour — giving the buyer power to set wages, affecting both wages and employment

Simple Monopsony

- In a monopsony, the firm faces the **market supply curve** for labour
 - Supply is upward sloping
 - Contrast to perfect competition where supply is **horizontal**
 - To hire more workers, the firm must pay a higher wage

! Marginal Cost of Labour

The firm's **marginal cost of labour is above the supply curve**

- To hire an additional worker, the firm must raise the wage for **all** workers
- $MC = \text{cost of additional worker} + \text{increase in wage paid to existing workers}$
- Analogous to how monopolists face MR below demand

Simple Monopsony

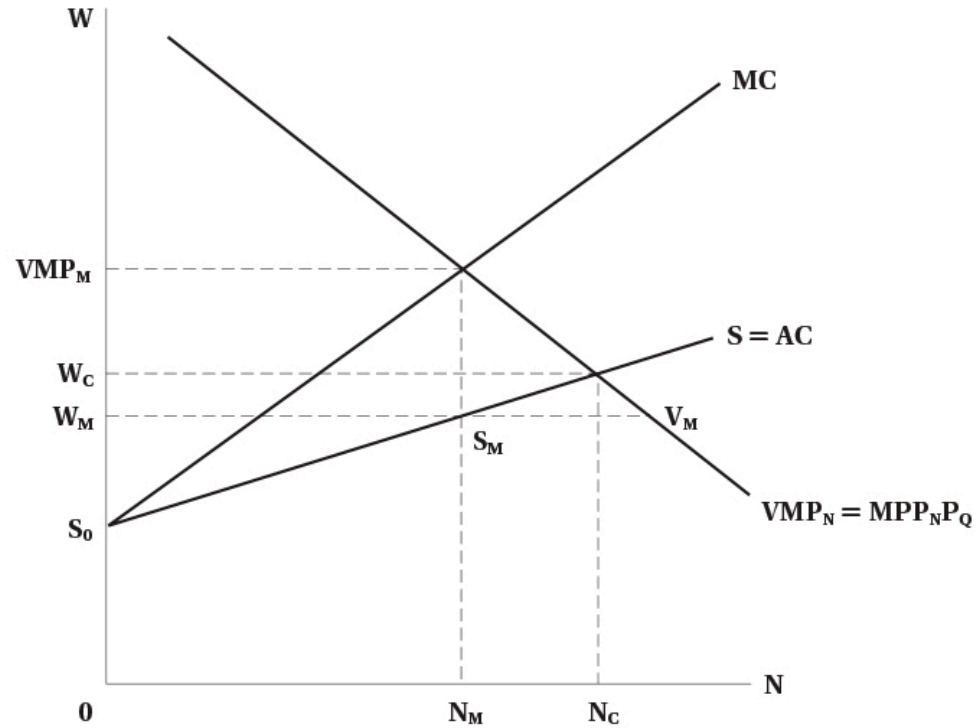


FIGURE 7.6 Monopsony

- Firm demand is its value of marginal product
- Faces upward sloping supply curve
- MC is above supply
- Optimal labour is where $MC = VMP$
- Wage comes from supply curve at that quantity of labour

Simple Monopsony

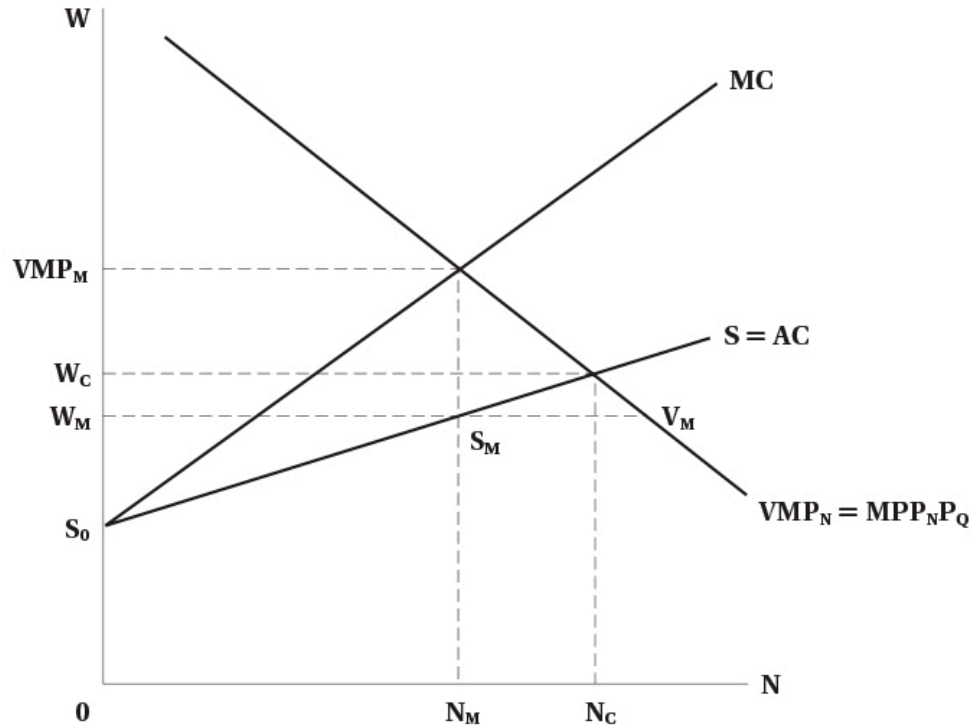


FIGURE 7.6 Monopsony

Compared to competitive equilibrium, monopsony results in:

- Wage below VMP
- Wage below competitive wage
- Employment below competitive level

The rectangle $(VMP_m - W_m) \times N_m$ represents **surplus transferred to the monopsonist** — the gap between what workers produce at the margin and what they are paid

Simple Monopsony

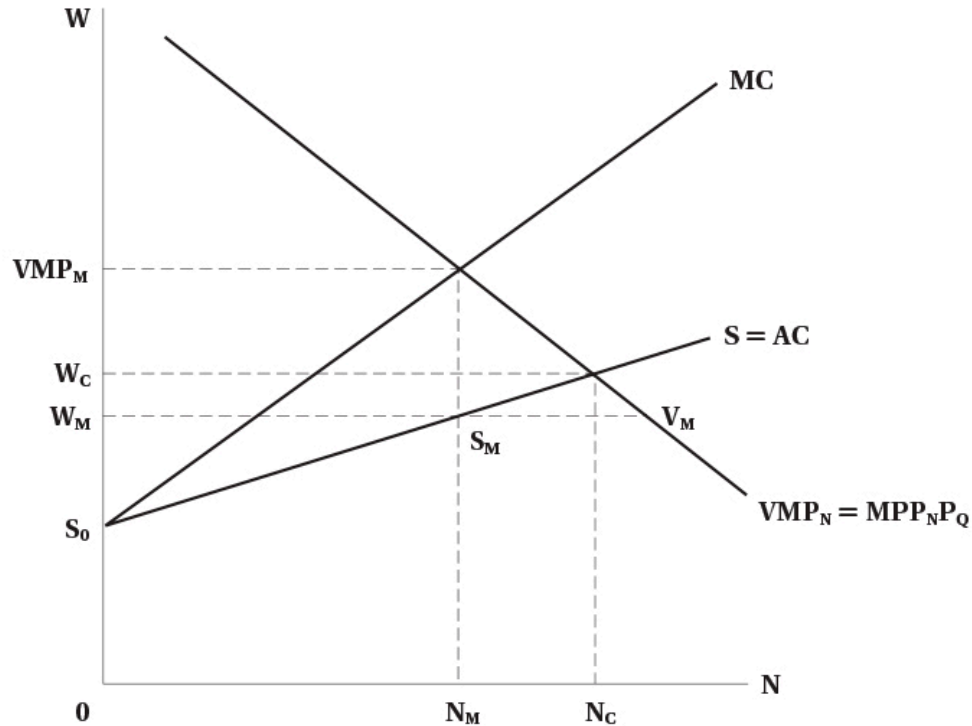


FIGURE 7.6 Monopsony

Monopsonies also leave **jobs vacant**:

- They restrict employment to maintain low wages
- This creates inefficiency in the labour market
- Vacancies are $V_m - S_m$

At wage W_m :

- VMP is above wage
- Firms would like to hire more
- But to hire, must raise wages — so it is optimal to not hire more

Think About It

What would happen to wages and employment if a monopsonist suddenly faced competition from a new firm entering the same labour market?

Characteristics of Monopsonies

Monopsonies tend to be:

- More prevalent in the short run
 - In the long run, workers can move to find other jobs
 - In the short run, frictions may prevent this
- Large relative to the labour market
 - Does not have to be a big firm, just large relative to the market
 - Example: a restaurant in a seasonal tourist town
- Associated with workers who have specialized skills
 - May naturally be few employers for specialized labour
 - Like academia, sports teams

Characteristics of Monopsonies

Note

Wages under monopsony are **lower than competitive equilibrium**, but not necessarily low in an absolute sense

- Example: sports players are not paid low wages, but market power makes them lower than they would be otherwise

Wage Differentiation

- In previous analysis, we assumed all workers are the same (single supply curve)
- This leaves some surplus for workers
 - There are workers willing to work for less than the equilibrium wage
- A firm can capture that surplus by differentiating wages
 - Paying different workers different wages based on their reservation wage

Wage Differentiation

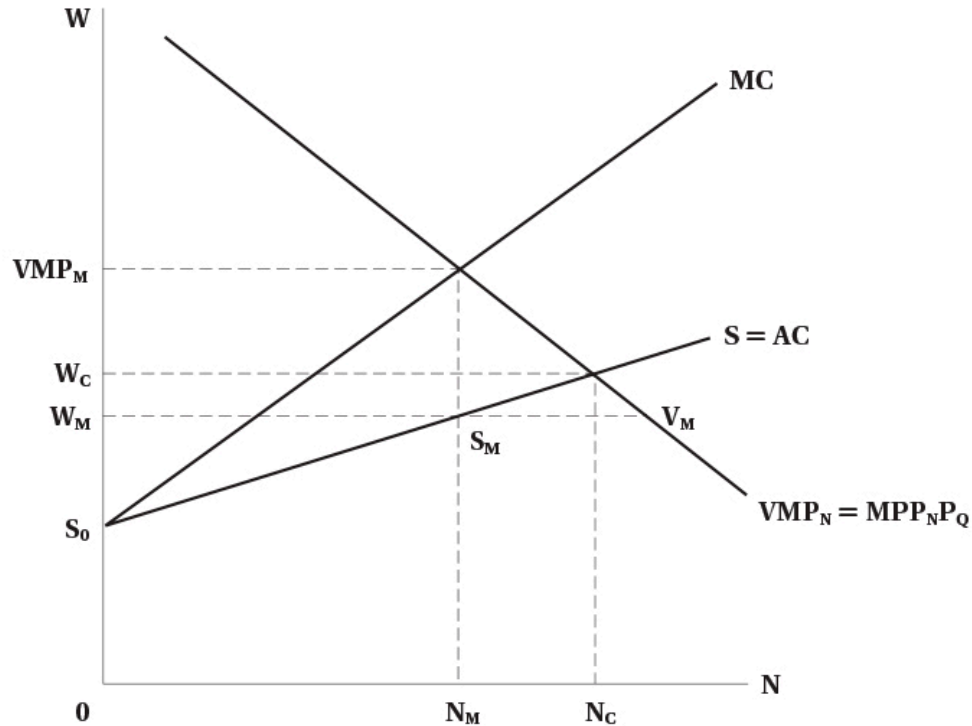


FIGURE 7.6 Monopsony

If the firm can pay different wages:

- It can pay each worker their **reservation wage**
- MC curve becomes equal to supply curve
- Firm will hire up to **competitive level**
 - It no longer has to increase wage for all workers to hire more

To do this, firms try to make wages secret:

- Prevent workers from knowing what others are paid
- Could use non-wage benefits for specific workers

Evidence of Monopsonies

Pro sports:

- Baseball: free agency increased wages significantly — increases competition among teams
- Hockey: not much evidence of monopsony power

Academia and nursing:

- Some evidence of monopsony power
- In Canada, a bargaining model may be more appropriate given these labour markets are unionized

Note

Some academics argue that monopsonies are relatively common — firms can set wages only if they have some market power, and evidence suggests this is widespread

Discussion

Why might monopsony power be more common than we think? Consider industries where workers have few alternative employers, even if the employer is not literally the only firm in town.

Minimum Wages

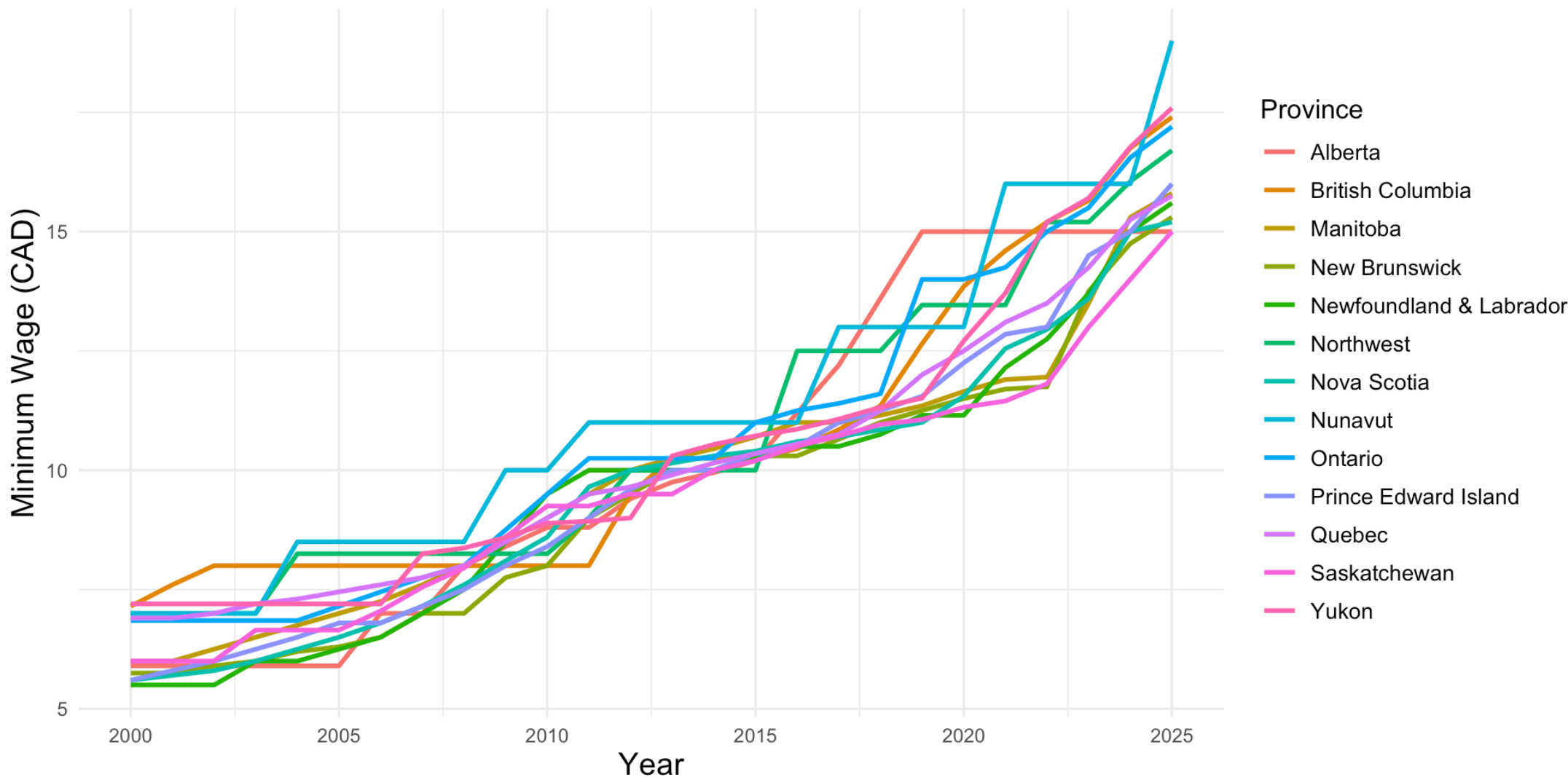
Introduction

! Minimum Wage

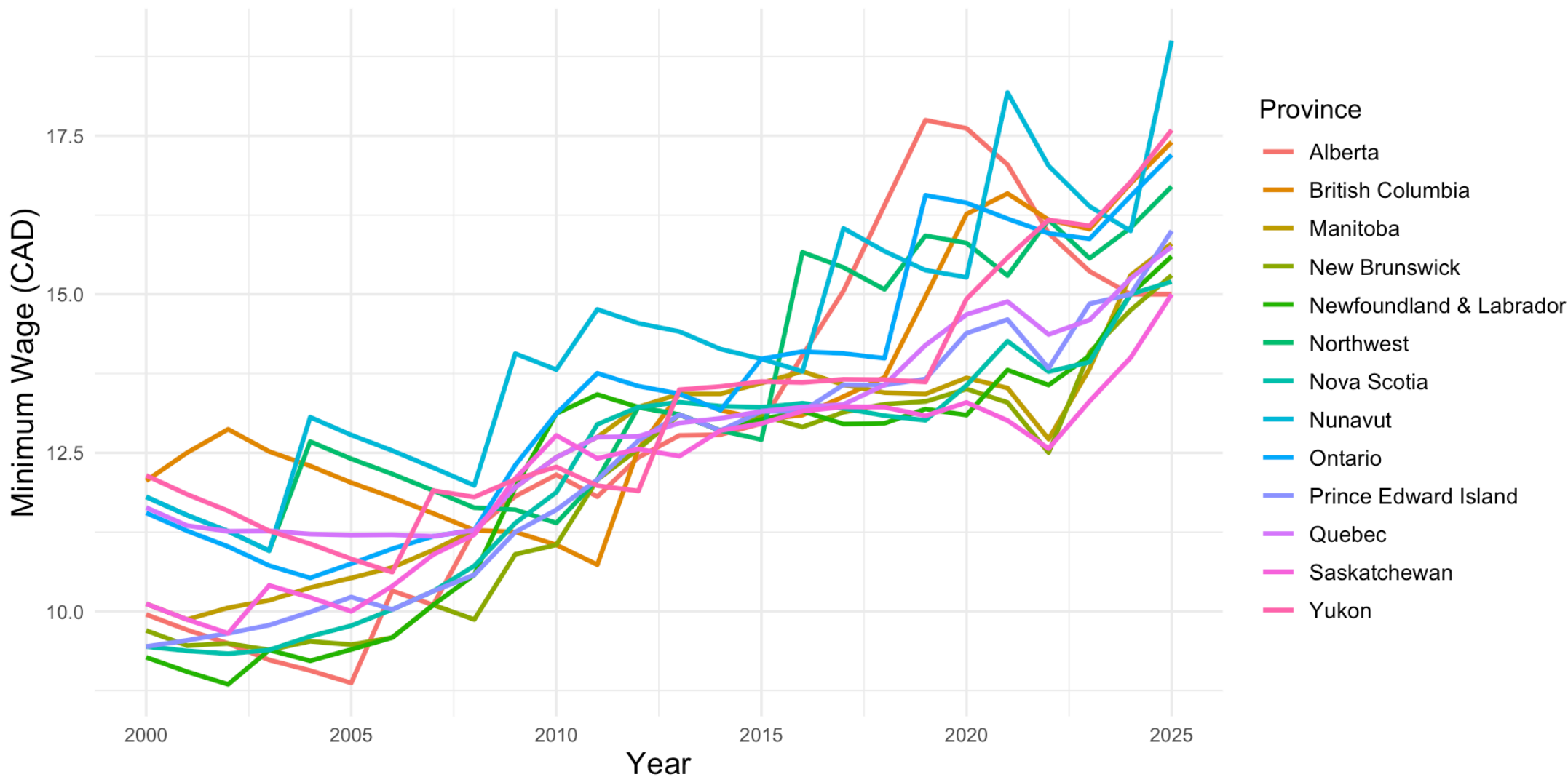
A **minimum wage** sets a legal floor on wages that employers must pay workers

- Intended to raise wages of low-wage workers, reduce poverty, increase standard of living
- Standard economic theory predicts that minimum wages lead to **unemployment**
 - Because they set a wage above the equilibrium wage
 - Quantity of labour supplied exceeds quantity demanded
- But statistical evidence is **much more mixed**

Nominal Minimum Wage by Province, 2000–2025



Real Minimum Wage by Province, 2000–2025



Minimum Wages in Canada and USA

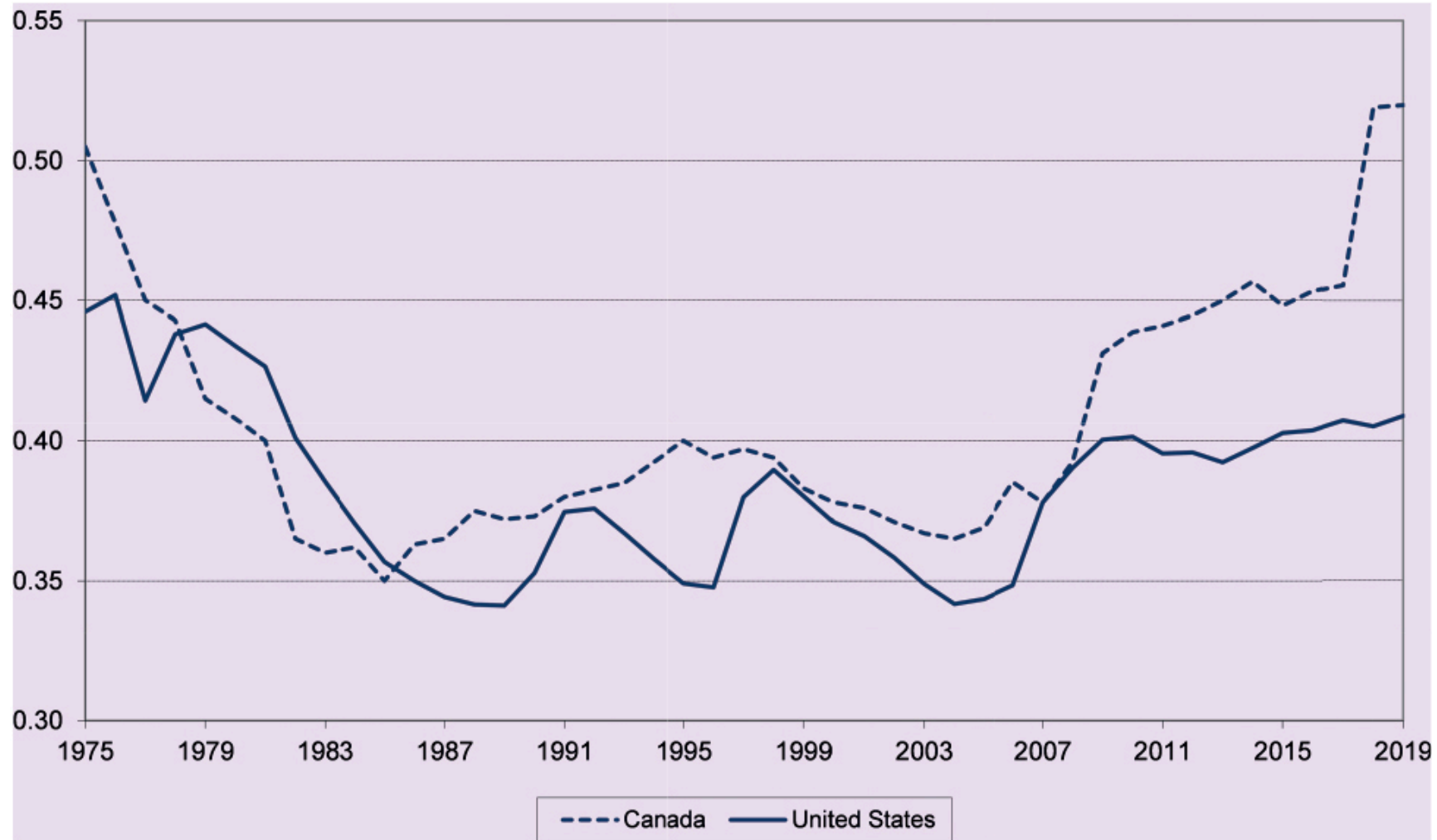


FIGURE 7.7 The Ratio of Minimum Wages to Average Wages, Canada and the United States, 1975–2019

Minimum Wages in Competitive Labour Market

In a **competitive labour market**:

- Market demand is downward sloping
- Market supply is upward sloping

A minimum wage acts as a **wage floor** set above the competitive equilibrium wage:

Warning

A binding minimum wage in a competitive market creates **excess supply of labour**, leading to unemployment

Minimum Wages in Competitive Labour Market

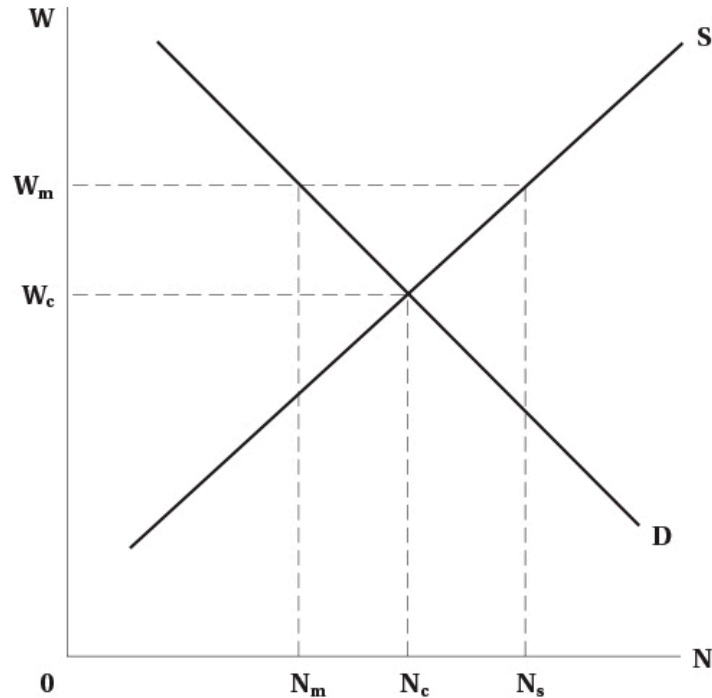


FIGURE 7.8 Minimum Wage

- Minimum wage W_m is above competitive level W_c
- Creates excess supply of labour
- Unemployment = $N_s - N_m$
 - Previously employed, now not: $N_c - N_m$
 - New entrants drawn in by higher wage: $N_s - N_c$

Minimum Wages in Competitive Labour Market

- Employment effects depend on **elasticity of demand**
 - More elastic demand → larger employment effects
- In the real world, industries with minimum wage workers tend to have elastic demands
 - Lots of substitutes for low-wage labour
 - Labour costs are a large share of total costs
- Nevertheless, we will see that minimum wages may not affect employment much in practice

Minimum Wages in Monopsony

Key Result

In a **monopsonistic** labour market, minimum wages can **increase both wages and employment**, depending on where the minimum wage is set

Minimum Wages in Monopsony

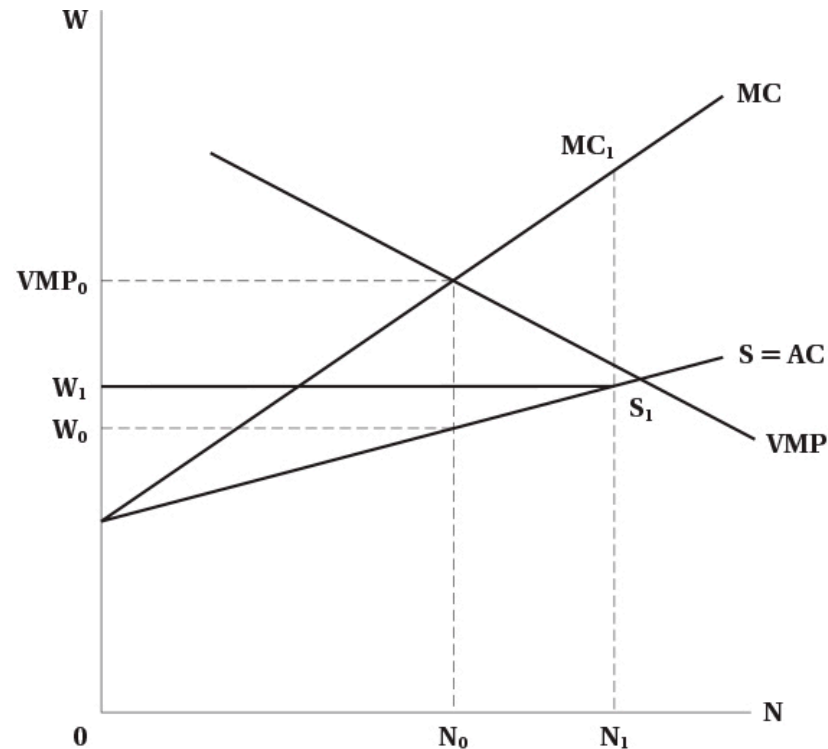


FIGURE 7.9 Monopsony and a Minimum Wage

- Prior to minimum wage: wage is W_0 , employment is N_0
- Minimum wage set above W_0 but below competitive wage
- MC curve = minimum wage up to S_1 , then jumps to previous MC curve
- Optimize where $MC = VMP$
- Employment increases to N_1

Minimum Wages in Monopsony

Why would minimum wages cause employment to **increase**?

- Previously, firm had to increase wages to attract more labour
- With minimum wage, it does not
 - Everyone is paid the same wage
 - Wage increases no longer inhibit employment growth
- Effectively, minimum wage lowers marginal cost

This may be partly why we do not see large employment effects of minimum wages

Note

Note that monopsonists do not like minimum wages — they reduce market power, increase costs, and reduce profits

Offsetting Factors for Minimum Wage

As usual, in models we hold all else equal — but in reality, many things change at once:

- Demand changes may accompany minimum wage increases
 - E.g., fiscal stimulus increasing demand for goods and labour
- Technology changes may follow
 - May change demand for labour
- Firms might choose to absorb costs
 - Reducing profits rather than cutting employment

Evidence

- Minimum wages are one of the most researched areas in labour economics
- Large amounts of evidence going back at least 50 years
- Early studies looked at time trends in employment and minimum wages
 - Subject to a lot of bias — things often trend together over time without direct causation
- Research in the early 1990s used better methods
 - Natural experiments
 - Difference-in-differences

Evidence

! Card and Krueger (1994)

- Looked at fast food restaurants in **New Jersey** (raised minimum wage) and Pennsylvania (did not)
- Used difference-in-differences to estimate employment effects
- Found **no negative employment effects** from the minimum wage increase
- Has been reevaluated several times — conclusion remains that employment effects are small or non-existent

Evidence

In Canada:

- Minimum wages **increase youth unemployment**
 - Studies looked at minimum wage changes across provinces in the 1990s
- Some Canadian studies look deeper at **labour market dynamics**
 - In any month, roughly 6% of employed people separate, 8% of non-employed people are hired
 - Evidence shows minimum wages reduce both hiring and separations
 - **Net effect on employment is approximately zero**

Evidence

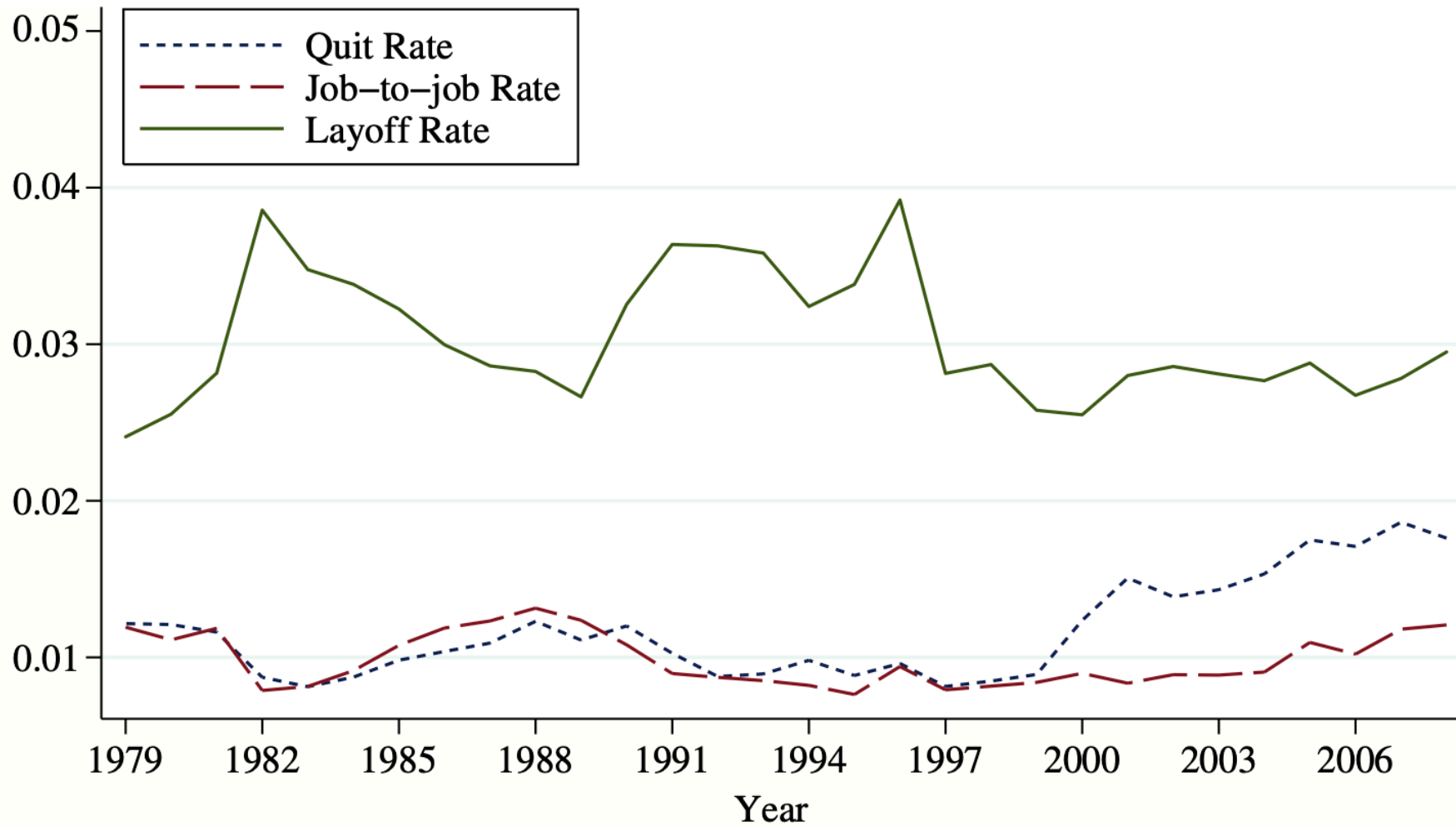


Fig. 4. *Layoff and Quit Rates, Low Skilled, 1979–2008*

Evidence

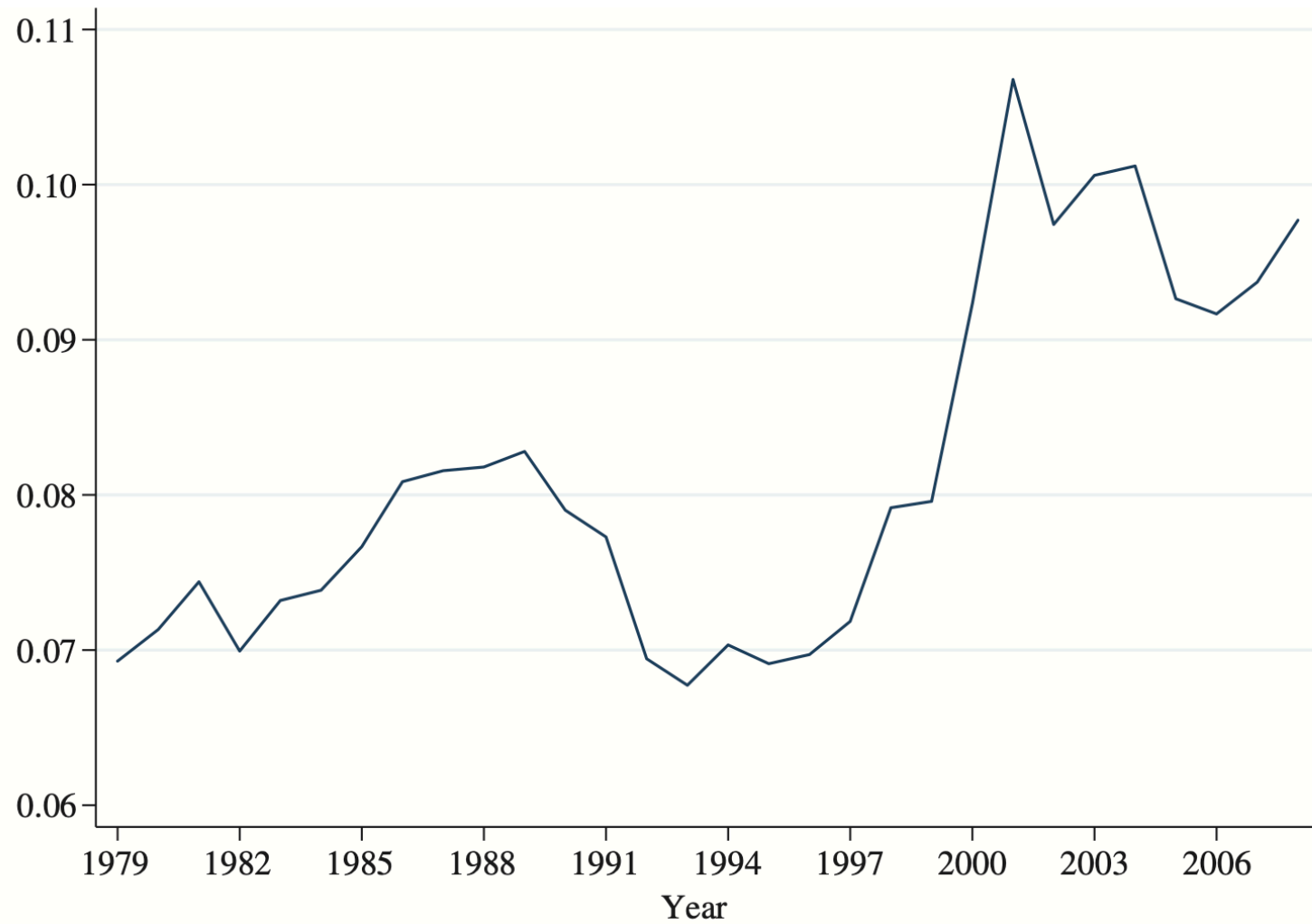


Fig. 5. *Hiring Rates, Low Skilled, 1979–2008*

Evidence

Warning

Minimum wages are generally **not effective at reducing poverty**

- Many poor individuals do not work
- Many people on minimum wages are not poor (e.g., youth living in high-income families)

Both Canadian and US studies show no link between minimum wages and poverty reduction

Discussion

If minimum wages do not effectively reduce poverty, what alternative policies might be more effective at helping low-income workers, and why?

Summary

Summary

Imperfectly competitive labour markets:

- A monopsonist faces an upward-sloping supply curve, so marginal cost of labour exceeds the wage
- The result is a wage and employment level below the competitive outcome
- A minimum wage reduces employment in a competitive market but can raise it under monopsony
- The empirical evidence on minimum wage employment effects is mixed and concentrated near zero